

Potentials and Fields

Problem 10.1

Show that the differential equations for V and $\mathbf{A}$

$$\begin{aligned}\nabla^2 V + \frac{\partial}{\partial t}(\nabla \cdot \mathbf{A}) &= -\frac{1}{\epsilon_0} \rho \\ \left(\nabla^2 \mathbf{A} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2} \right) - \nabla \left(\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} \right) &= -\mu_0 \mathbf{J}\end{aligned}$$

can be written in the more symmetrical form

$$\begin{aligned}\square^2 V + \frac{\partial L}{\partial t} &= -\frac{1}{\epsilon_0} \rho \\ \square^2 \mathbf{A} - \nabla L &= -\mu_0 \mathbf{J}\end{aligned}$$

where

$$\square^2 \equiv \nabla^2 - \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2} \quad \text{and} \quad L \equiv \nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t}$$

Solution:

Put the definitions of $\square^2$ and L into the equations for V :

$$\begin{aligned}\left(\nabla^2 - \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2} \right) V + \frac{\partial}{\partial t} \left(\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} \right) &= \\ = \nabla^2 V - \mu_0 \epsilon_0 \frac{\partial^2 V}{\partial t^2} + \frac{\partial(\nabla \cdot \mathbf{A})}{\partial t} + \mu_0 \epsilon_0 \frac{\partial^2 V}{\partial t^2} &= \\ = \nabla^2 V + \frac{\partial}{\partial t}(\nabla \cdot \mathbf{A}) &= -\frac{1}{\epsilon_0} \rho \quad \# \end{aligned}$$

and for $\mathbf{A}$:

$$\begin{aligned} & \left(\nabla^2 - \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2} \right) \mathbf{A} - \nabla \left(\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} \right) = \\ & = \left(\nabla^2 \mathbf{A} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2} \right) - \nabla \left(\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} \right) = -\mu_0 \mathbf{J} \quad \# \end{aligned}$$

□

Problem 10.2

Problem 10.3

(a) Find the fields, and the charge and current distributions, corresponding to

$$V(\mathbf{r}, t) = 0, \quad \mathbf{A}(\mathbf{r}, t) = -\frac{1}{4\pi\epsilon_0} \frac{qt}{r^2} \hat{\mathbf{r}}$$

(b) Use the gauge function $\lambda = -(1/4\pi\epsilon_0)(qt/r)$ to transform the potentials, and comment on the result.

Solution:

The $\mathbf{E}$ and $\mathbf{B}$ fields are

$$\begin{aligned} \mathbf{E} &= -\nabla V - \frac{\partial \mathbf{A}}{\partial t} \\ \mathbf{B} &= \nabla \times \mathbf{A} \end{aligned}$$

The derivative of $\mathbf{A}$ with respect to time is

$$\frac{\partial \mathbf{A}}{\partial t} = -\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}},$$

while the curl of $\mathbf{A}$ in spherical coordinates is

$$\frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta A_\phi) - \frac{\partial A_\theta}{\partial \phi} \right] \hat{\mathbf{r}} + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r} (r v_\phi) \right] \hat{\boldsymbol{\theta}} + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right] \hat{\boldsymbol{\phi}}.$$

Since $A_\phi = A_\theta = 0$, and $\frac{\partial A_r}{\partial \phi} = \frac{\partial A_r}{\partial \theta} = 0$, the curl of $\mathbf{A}$ is zero.

(a)

$$\begin{aligned} \mathbf{E} &= -\frac{\partial \mathbf{A}}{\partial t} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}} \\ \mathbf{B} &= \nabla \times \mathbf{A} = 0 \end{aligned}$$

(b) Using the gauge function, V and $\mathbf{A}$ are transformed into

$$\begin{aligned} V'(\mathbf{r}, t) &= V(\mathbf{r}, t) - \frac{\partial \lambda}{\partial t} \\ \mathbf{A}'(\mathbf{r}, t) &= \mathbf{A}(\mathbf{r}, t) + \nabla \lambda \end{aligned}$$

Writing out the derivative and gradient of λ

$$\begin{aligned} \frac{\partial \lambda}{\partial t} &= -\frac{1}{4\pi\epsilon_0} \frac{q}{r} \\ \nabla \lambda &= \frac{1}{4\pi\epsilon_0} \frac{qt}{r^2} \hat{\mathbf{r}} \end{aligned}$$

gives the transformed potentials

$$\begin{aligned} V'(\mathbf{r}, t) &= \frac{1}{4\pi\epsilon_0} \frac{q}{r} \\ \mathbf{A}'(\mathbf{r}, t) &= -\frac{1}{4\pi\epsilon_0} \frac{qt}{r^2} \hat{\mathbf{r}} + \frac{1}{4\pi\epsilon_0} \frac{qt}{r^2} = 0 \end{aligned}$$

$\mathbf{E}$ is now given by $-\nabla V$ (instead of by $-\frac{\partial \mathbf{A}}{\partial t}$), while $\mathbf{B}$ is again 0, this time directly because $\nabla \times \mathbf{A} = \nabla \times 0 = 0$.

$$\begin{aligned} \mathbf{E} &= -\nabla V = -\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}} \\ \mathbf{B} &= \nabla \times \mathbf{A} = 0 \end{aligned}$$

The result is exactly the same as in (a).

□

Problem 10.4

Suppose $V = 0$ and $\mathbf{A} = A_0 \sin(kx - \omega t) \hat{\mathbf{y}}$, where A_0, ω , and k are constants. Find $\mathbf{E}$ and $\mathbf{B}$, and check that they satisfy Maxwell's equations in vacuum. What conditions must you impose on ω and k ?

Solution:

Maxwell's equations in vacuum are

$$\begin{aligned}
\text{(i)} \quad \nabla \cdot \mathbf{E} &= \frac{1}{\epsilon_0} \rho & \text{(iii)} \quad \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\
\text{(ii)} \quad \nabla \cdot \mathbf{B} &= 0 & \text{(iv)} \quad \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}
\end{aligned}$$

where as before we set

$$\begin{aligned}
\mathbf{E} &= -\nabla V - \frac{\partial \mathbf{A}}{\partial t} \\
\mathbf{B} &= \nabla \times \mathbf{A}
\end{aligned}$$

This gives

$$\begin{aligned}
\mathbf{E} &= -\frac{\partial \mathbf{A}}{\partial t} = -\omega A_0 \cos(kx - \omega t) \hat{\mathbf{y}} \\
\mathbf{B} &= \frac{\partial A_y}{\partial x} \hat{\mathbf{z}} = k A_0 \cos(kx - \omega t) \hat{\mathbf{z}}
\end{aligned}$$

Since $\rho = 0$, this directly gives $\nabla \cdot \mathbf{E} = 0$. (ii) gives $\nabla \cdot \mathbf{B} = 0$.

From (iv) we have the relation, since the current density $\mathbf{J} = 0$,

$$\begin{aligned}
\nabla \times \mathbf{B} &= \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \\
&\downarrow \\
\nabla \times \mathbf{B} &= -\frac{\partial B_z}{\partial x} \hat{\mathbf{y}} = k^2 A_0 \sin(kx - \omega t) \hat{\mathbf{y}} \\
\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} &= \mu_0 \epsilon_0 \omega^2 A_0 \sin(kx - \omega t) \hat{\mathbf{y}}
\end{aligned}$$

From this we see that $k^2 = \mu_0 \epsilon_0 \omega^2 \rightarrow k = \sqrt{\mu_0 \epsilon_0} \omega \rightarrow k = \frac{\omega}{c}$

From (iii) we have the relation

$$\begin{aligned}
\nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\
&\downarrow \\
\nabla \times \mathbf{E} &= \frac{\partial E_y}{\partial x} \hat{\mathbf{z}} = \omega k A_0 \sin(kx - \omega t) \hat{\mathbf{z}} \\
-\frac{\partial \mathbf{B}}{\partial t} &= \omega k A_0 \sin(kx - \omega t) \hat{\mathbf{z}}
\end{aligned}$$

Equations (i)-(iv) are all verified with the condition $k = \frac{\omega}{c}$.

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